

GenAI · Education

Research · Learning

AI Literature Review Assistant: Teaching Students How Research Works

A human-in-the-loop research-learning companion that guides university students from topic formulation to evidence-grounded synthesis. Working prototype for the GenAI for Education Hackathon 2026.

Open Challenge

EUI GenAI for Education Hackathon · Student Track



THE RESEARCH PROBLEM

Literature Review Is More Than Finding Papers

Students are expected to conduct research, but the process is fragmented across search engines, PDFs, notes, spreadsheets, and chatbots.

What students must reason through

- Turn a broad topic into useful searches
- Decide which papers are truly relevant
- Understand methods, findings, and limitations
- Compare studies and distinguish evidence from interpretation

Where the workflow breaks down

- Scattered notes and lost context
- Manual cross-paper comparison
- Uncertain research-gap claims
- Weak links between claims and sources

Students Need to Learn the Research Process

The challenge is not simply summarization. Students need structured support for research reasoning—not an answer generator.

01

Generic chatbot

Ask a question and receive an answer, often without a visible research workflow.

02

Research learning workflow

Topic → search → select → analyze → compare → synthesize → validate.

03

Student stays involved

The learner reviews papers, evaluates claims, and makes important decisions.



A Guided, Evidence-Grounded Research Companion

AI Literature Review Assistant turns literature review into a visible, structured learning workflow.

1

Discover

Generate research queries and retrieve candidate papers from arXiv.

2

Understand

Analyze approved papers for problems, methods, findings, limitations, and relevance.

3

Synthesize

Compare studies, identify candidate gaps, validate evidence, and support follow-up questions.





HUMAN-IN-THE-LOOP WORKFLOW

AI Assists. The Student Decides.

The prototype pauses before deep analysis so students can review and select the papers that enter their literature review.



Generate and retrieve

Turn a topic into academic queries and retrieve candidate papers from arXiv.



Approve evidence

The student reviews relevance and decides which papers should be analyzed.



Analyze in parallel

Selected papers are structured into problems, methods, findings, limitations, and relevance.



Validate the synthesis

Candidate gaps and important claims are checked against the reviewed evidence.

Use GenAI for Reasoning. Use Deterministic Systems for Control.

The architecture is intentionally hybrid: language-heavy tasks use generative reasoning, while retrieval and validation use predictable controls.

01

GenAI reasoning

Query generation, relevance interpretation, methodology analysis, findings interpretation, synthesis, and gap reasoning.

02

Deterministic control

Retrieval, deduplication, existence checks, structural validation, and citation-set validation.

03

Workflow orchestration

LangGraph nodes, interrupts and resume, checkpointed state, and parallel paper analysis.

04

Current status

Working prototype / MVP using Groq-based LLM calls, arXiv retrieval, and Markdown report rendering.

Trust by design

Important student decisions remain visible, and claims are kept connected to the evidence reviewed.

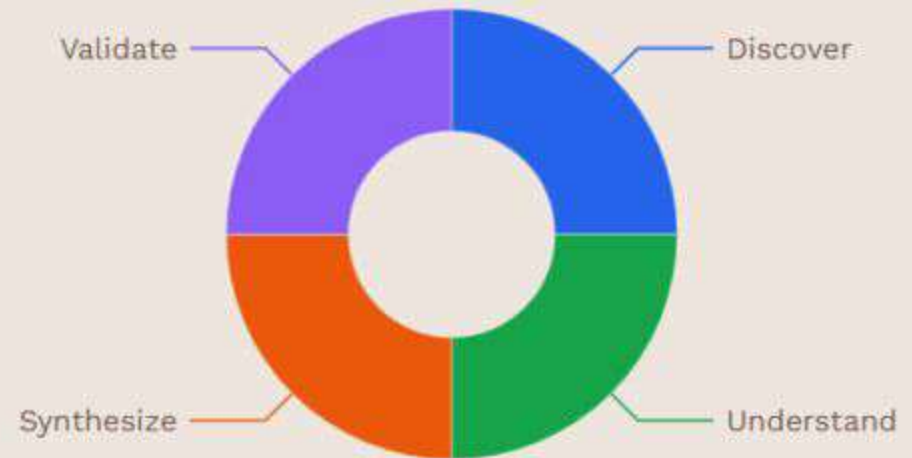
Built for learning

The system exposes the reasoning stages instead of hiding them behind a single generated answer.



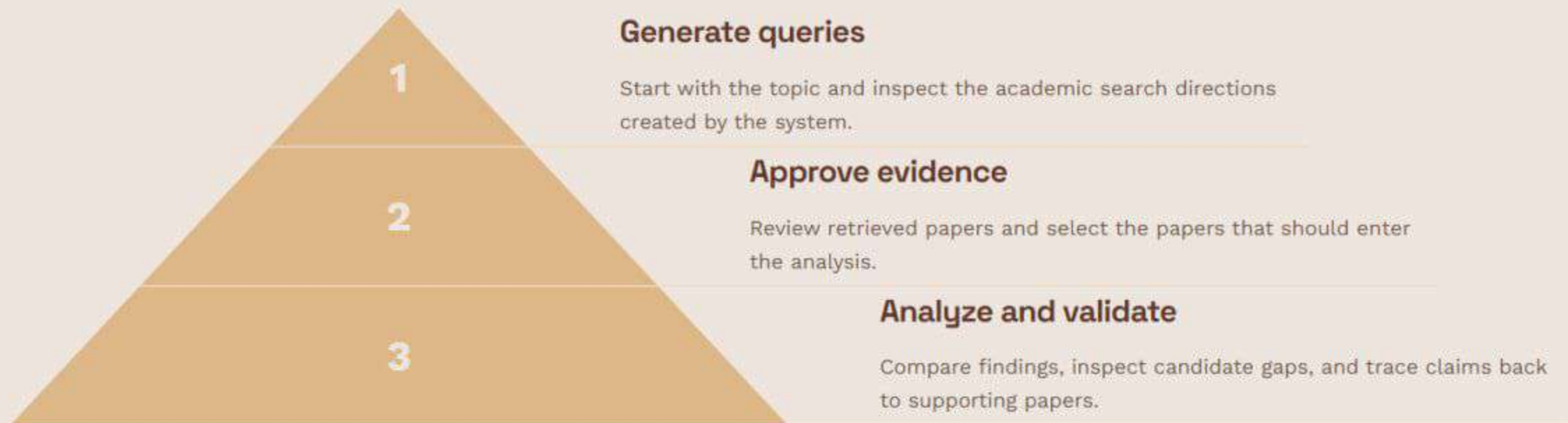
From Scattered Research Work to Guided Research Practice

The goal is not merely faster literature reviews. It is helping students understand how a defensible literature review is built.



One Research Question, One Complete Workflow

Demo topic: **“Early Failure Prediction in Multi-Agent Systems”**. The audience should see a research process—not a magic answer.



Visible progress

Generate queries → review papers →
approve evidence → analyze → compare
→ validate.



Student checkpoint

The AI proposes candidates, but the student decides what belongs in the review.



Evidence continuity

Follow-up questions preserve the context of the completed literature-review round.

AI Should Not Hide the Research Process—It Should Make It Learnable

The student remains the researcher. AI provides structure, reasoning support, and evidence organization.

AI Literature Review Assistant is a human-in-the-loop research-learning companion that guides students from topic formulation to evidence-validated synthesis.

Future direction: expand beyond arXiv, support more disciplines and Arabic academic work, add personalized feedback, and evaluate learning outcomes with real students.

